

GIS for veterinary disease surveillance and control

John Grewar



Daily Program – Day 1-3*

**subject to change*

- 08:00 – 10:00
 - Session 1
 - Introduction by SAVA and SASVEPM
 - Facilitator presentations and in course tasks
- 10:00 – 10:30
 - Tea
- 10:30 – 12:00
 - Session 2
 - Facilitator presentations and in course tasks
- 13:00 – 15:00
 - Session 3
 - Facilitator presentations and in course tasks
- 15:00 – 15:30
 - Tea
- 15:30 – 16:45
 - Session 4
 - Facilitator presentations and in course tasks
- 16:45 – 17:00
 - Wrap of Day - all

<https://us06web.zoom.us/j/85011282025?pwd=vFGQvaLWb95XL1YoAcnDL4UM6tLaYv.1>

Meeting ID: 850 1128 2025
Passcode: 677070



Daily Program – 3

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 - Session 2
 - Facilitator presentations and in course tasks
- 13:00 – 16:00
 - Session 3
 - Assessment session with access to Facilitator
- 16:00 – 16:30
 - Session 3
- 16:30 – 17:00
 - Wrap of Day – all and the Assessment of course

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Ice Breaker

Please insert three emoji's into the Zoom Chat which best describes you work in regard to mapping of diseases or disease control/surveillance



Outline

- Section 1: Introduction to Geographic Information Systems
 - GIS Data
 - Vector
 - Raster
 - Coordinate reference systems
 - Spatial Analysis
- Section 2: Introduction to qGIS and its interface
 - Menus, Browser, Layers, Toolbox
 - Attribute tables
- Section 3: Basic qGIS operations
 - Adding vector layers
 - CSV, Shapefiles, GeoPackage
 - Navigating qGIS and symbolising data
 - Colour, Category, Quantity
 - Labelling data
 - Vector classification
 - Field calculator
 - Choropleth maps
- Section 4: Exporting Maps for reports/publication with the Print layout
- Section 5: Vector analysis
 - Filtering
 - Buffers
 - Interpolation
- Final assessment – AHS outbreak



Files for course

- https://drive.google.com/file/d/1n6h_LvHQ9iv850jT1FdUnW6g56VayE75/view?usp=drive_link



Assignments

- Your assignments for the project should be uploaded at:
- <https://five.epicollect.net/project/gis-assessments>
 - For this you will need to login at EpiCollect5 (<https://five.epicollect.net/>) and make an account beforehand – it is free



SECTION 1

- Introduction to Geographic Information Systems



Introduction to GIS and some basic concepts

- A Geographic Information System contains
 - Spatial data
 - Computer hardware
 - Computer software
 - qGIS
 - ArcGIS
 - R/Python



Typical uses of GIS in Animal Health

- **Mapping**
- **Spatial Analysis**
 - Analyze spatial relationships, conduct site suitability analysis, calculate distances
- **Data Conversion**
 - Convert geospatial data between different formats
- **Database Integration**
- **Remote Sensing**
- **Data Visualization**

Part 1: Introduction to QGIS

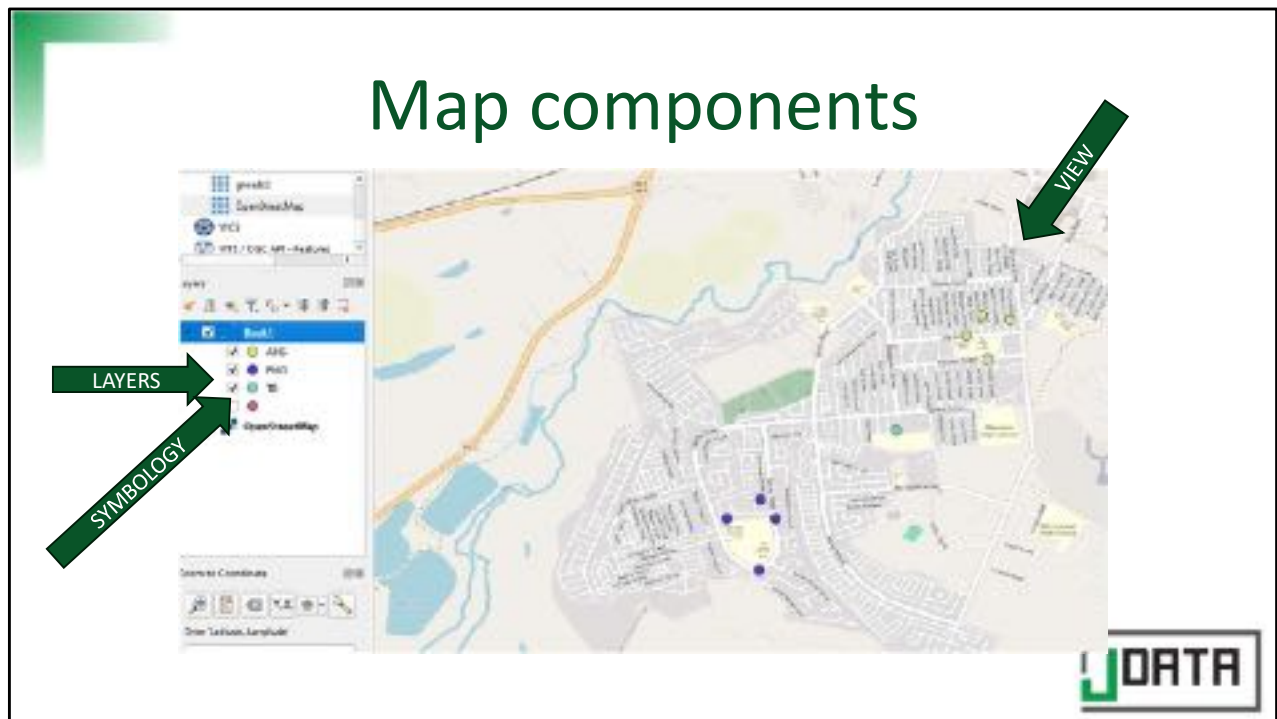


GIS Data

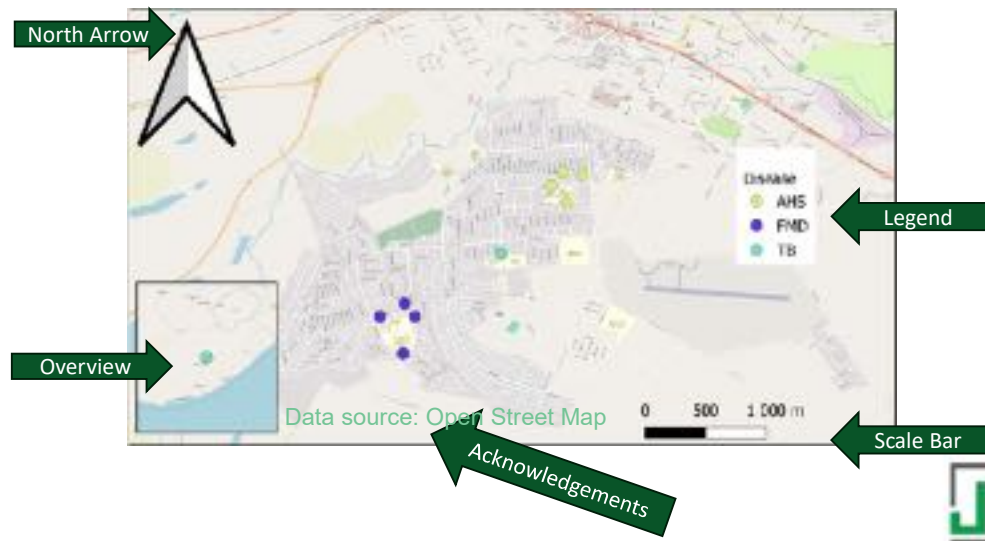
Longitude	Latitude	Disease	Date
26.870436	-31.909519	AHS	11/12/2024
26.868682	-31.909259	AHS	24/12/2024
26.867707	-31.910494	AHS	22/01/2025
26.854908	-31.920759	FMD	11/01/2025
26.855817	-31.921929	FMD	26/01/2025
26.852764	-31.921929	FMD	09/02/2025
26.854778	-31.925112	FMD	22/02/2025
26.869072	-31.911988	AHS	02/02/2025
26.863354	-31.916406	TB	26/02/2025



Map components

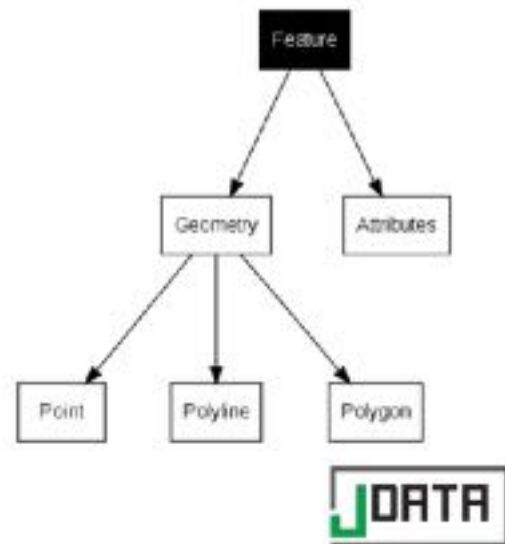


Map components



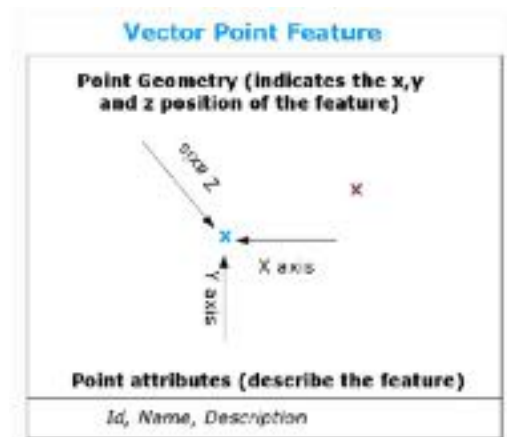
Vector data

- Point
 - Holding
- Polyline
 - River, Road
- Polygon
 - Dams, Provinces



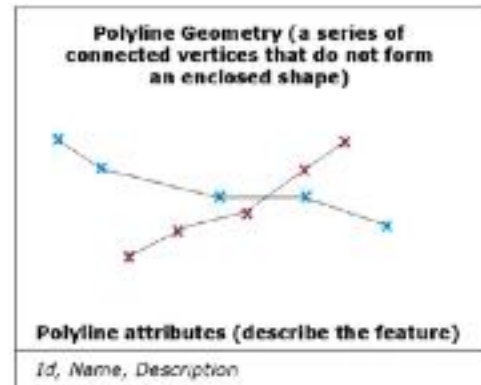
Vector data - Points

- A point feature is described by its X, Y and optionally Z coordinate. The point attributes describe the point
- Every point can have multiple **attributes**



Vector data - Polylines

- A polyline is a sequence of joined vertices. Each vertex has an X, Y (and optionally Z) coordinate



Vector data - Polygons

- A polygon, like a polyline, is a sequence of vertices. However, in a polygon, the first and last vertices are always at the same position

